Steel City (Major Urban / Large-Scale Combined Arms)

Executive Summary

Steel City is a massive **urban-industrial megabattlefield** built for **large-scale combined arms warfare**, integrating armor, infantry, drones, artillery, air support, and autonomous units across dense city blocks and heavy industrial zones. The terrain features high-rise clusters, steelworks, rail corridors, elevated freeways, cargo terminals, fortified intersections, and multi-level combat spaces. The map emphasizes Obsidian Protocol's autonomous coordination across multiple fronts, sensor-network routing, comms-relay warfare, and logistics under dense urban constraints. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of major urban terrain. Key features include multi-tier street networks, industrial choke zones, rooftop combat, and large-scale combined arms engagements

Design Goals & Gameplay

Large-Scale Combined Arms Warfare

Steel City supports simultaneous armor pushes, infantry assaults, drone reconnaissance, artillery strikes, and air support. Multi-level terrain enables layered combat.

Urban-Industrial Hybrid Combat

High-rise districts provide vertical combat; industrial zones provide long-range fire lanes; rail corridors enable flanking; elevated freeways create multi-tier movement.

Autonomy & Multi-Front Coordination

Autonomous units coordinate across multiple sectors, selecting optimal fronts to reinforce or breach based on sensor data and objective priority.

Sensor & Information Warfare

Smoke, dust, steelworks heat, and dense building layouts distort sensors. Relay towers on rooftops restore clarity. Destroying relays creates blind pockets across the city grid.

Logistics & Resources

Cargo terminals, steelworks, rail yards, and freeway checkpoints serve as resupply nodes. Convoys must navigate unpredictable street networks and avoid ambush-prone industrial corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between heavy armor, air support, artillery, or autonomous infantry squads. Blue deploys from the southwest industrial yard; Red deploys from the northeast high-rise command tower.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² major urban zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Industrial Yard Command) — (0,0), 150x150 m Steelworks, cargo cranes, armored staging area

Red HQ (High-Rise Command Tower) — (1000,1000), 150x150 m Skyscraper command node with radar mast and AA emplacements

Central Steel District — (500,450), 300x300 m Factories, warehouses, smokestacks, long-range industrial fire lanes

Cargo Terminal & Rail Hub — (250,800), 150x150 m Rail spurs, cargo cars, loading cranes, flanking routes

High-Rise Cluster — (800,300), 200x200 m Multi-level combat, rooftops, interior corridors

Elevated Freeway Spine — (0,450 → 400,900) Two-tier freeway; critical movement corridor and chokepoint.

Lower Street Grid — (300,0 → 450,350) Dense urban streets; ideal for infantry and autonomous squads

Rooftop Relay A — (650,150), +260 m Long-range sensor vantage point.

Rooftop Relay B — (150,650), +280 m Comms relay tower and drone hub

Neutral Forward Outpost — (500,950) Drone resupply zone

Chokepoints & Sightlines

Elevated Freeway Spine

Major chokepoint; controlling it determines north-south movement.

Steel District Fire Lanes

Long-range industrial corridors ideal for armor and artillery.

High-Rise Cluster

Vertical combat; rooftops provide overwatch and sensor dominance.

Lower Street Grid

Close-range combat; ideal for autonomous infantry and ambushes.

Sightlines

Rooftop Relay A provides long-range sensor coverage (~350 m). Smoke and industrial heat create fog-of-war pockets

Deployment Zones

Blue Deployment: SW industrial yard + armored staging

Red Deployment: NE high-rise command tower + AA emplacements.

Neutral Forward Zone: Outpost at (500,950) for drone drops

Assets & Environment

Terrain

High-rises, industrial zones, rail yards, freeways, street grids.

Vegetation

Sparse trees, planters, park strips.

Buildings & Props

Military: Watchtowers, AA guns, radar dishes. Industrial: Factories, cranes, pipelines, generators. Urban: Apartments, offices, shops, parking structures. Logistics: Trucks, crates, fuel drums, drones.

Effects

Smoke, sparks, dust, fog, heat shimmer.

Audio

Machinery, traffic hum, distant artillery, radio chatter.

LOD & Budget

Asset Cate	**Count**	**Tri/Item**	**Total Tris**
High-Rise	3	200k	600k
Industrial	4	80k	320k
Buildings/	15	50k	750k
Roads/Fre	4	30k	120k
Vegetatio	120	2k	240k
Props	50	10k	500k
Total	—	—	**~2.53M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out high-rises, steel district, freeway, AR anchor test
2. **Core Systems — Month 2** NavMesh, combined-arms autonomy behaviors
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (smoke, sparks, heat shimmer)
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation

Testing Plan

Autonomy

Verify multi-front coordination, combined-arms logic, route selection.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, combined-arms behavior, deployment budget enforcement, anchor drift, sensor accuracy.